

Pick one discrete and one continuous distribution we have covered, and try and find an interesting application of it. If you can find one from within AI research or practice that would be great!

Bernoulli distribution used in LLM benchmarking.

Each pass or fail test score is considered a Bernoulli variable as there are 2 outcomes pass or fail.

I read about this in the paper: [Adding Error Bars to Evals \(Miller 2024\)](#)

From our lessons, the **probability mass function** is:

$$P(X = 1) = p, \quad P(X = 0) = (1 - p)$$

This gives an expectation $E(X) = p$ of and a variance of $Var(X) = p(1 - p)$

A benchmark score is simply the mean of n such tests so the sum of all the scores divided by n so:

$$\bar{X} = \frac{\text{Total scores } (S)}{n} \text{ so then } Var(\bar{X}) = \frac{1}{n^2} \cdot n \cdot Var(x) = \frac{Var(x)}{n}$$

The standard deviation or rather standard error is then the square root of $Var(\bar{X})$ which Miller defines as the following:

$$SE_{Bernoulli} = \sqrt{\frac{\bar{s}(1-\bar{s})}{n}}$$

Where $\bar{S}$ is the proportion of questions answered correctly (accuracy), which is an estimate for the model's true probability.

The SE helps us to understand how much model performance would vary if the evaluation was repeated either with the same set of questions or a new set. If we look at Table 2 we can see that for the HumanEval case the SE is ~ 3% but the difference in the model performance is also ~ 3% so we cannot conclude which model is better than the other. We can also see that for the MATH question set of 5000 questions the SE was much lower, 0.7% while HumanEval only had 164, so the precision of the benchmark is very much dependent on the number of questions it has.

Miller gives the following equation for a score on a particular evaluation question:

$$s_i = x_i + \epsilon_i$$

Where:

- s_i the score on this particular question, which is the **random variable** (we had previously called X).
- x_i is the **probability** the question is answered correctly, that we had in our lessons called p .
- ϵ_i is the difference between the model's score on this question and the score it would get on average. This comes from the fact that models are non-deterministic so any single attempt can differ from the average and this difference is the noise.

Uniform distribution used for question difficulty in LLM benchmarking

Sticking with the same paper [Adding Error Bars to Evals \(Miller 2024\)](#) . Miller explores ways to reduce the standard error. He gives variance as:

$$\text{Var}(\bar{\mu}) = \frac{\text{Var}(s)}{n} = \frac{\text{Var}(x) + E[\sigma_i^2]}{n}$$

Where:

- $\text{Var}(x)$ is the variance in the difficulty of different questions
- $E[\sigma_i^2]$ comes from the same question being re-run and giving different answers (non-deterministic nature of LLMs)

Miller uses the Uniform distribution to model question difficulty, he assumes that question difficulty is uniformly distributed $x \sim U[0, 1]$. There are some questions the model always passes and some it always fails. From our lesson Exercise $E(X) = \frac{a+b}{2}$. I attempted to calculate the variance for the Uniform distribution in overleaf:

$$\begin{aligned} E(X^2) &= \int_{-\infty}^{\infty} x^2 f(x) dx \\ &= \int_a^b x^2 \frac{1}{b-a} dx \\ &= \left[\frac{x^3}{3(b-a)} \right]_a^b \\ &= \frac{b^3 - a^3}{3(b-a)} \\ &= \frac{a^2 + ab + b^2}{3} \end{aligned}$$

$$\begin{aligned} \text{Var}(X) &= E(X^2) - [E(X)]^2 \\ &= \frac{a^2 + ab + b^2}{3} - \left(\frac{a+b}{2} \right)^2 \\ &= \frac{4(a^2 + ab + b^2) - 3(a+b)^2}{12} \\ &= \frac{4a^2 + 4ab + 4b^2 - 3a^2 - 6ab - 3b^2}{12} \\ &= \frac{a^2 - 2ab + b^2}{12} \\ &= \frac{(b-a)^2}{12} \end{aligned}$$

So for this paper with upper bound $b=1$ and lower bound $a=0$:

$$Var(X) = \frac{(1-0)^2}{12} = \frac{1}{12}$$

Since $Var(X)$ comes from the questions themselves and not from the model, it sets a limit on how much the standard error can be improved.

He also computes $E[\sigma_i^2]$:

For one question with difficulty x_i the score is Bernoulli so the variance is $p(1-p)$ and so the standard deviation squared is:

$$\sigma_i^2 = Var(\sigma_i) = x_i(1 - x_i)$$

We need the average across all the questions where difficulty will vary from question to question:

$$E[\sigma_i^2] = \int_0^1 x(1 - x) \cdot f(x) dx$$

$f(x) = \frac{1}{b-a}$ for Uniform probability density function, $f(x) = \frac{1}{1-0} = 1$. So:

$$\begin{aligned} E[\sigma_i^2] &= \int_0^1 x(1 - x)dx = \int_0^1 (x - x^2)dx \\ &= \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 \\ &= \left(\frac{1}{2} - \frac{1}{3} \right) - 0 \\ &= \frac{1}{6} \end{aligned}$$

So total variance is $\frac{1}{6} + \frac{1}{12} = \frac{1}{4}$. So running the same questions several times on the model could reduce the variance by at most two thirds of the total variance.

Then, find one discrete and one continuous distribution which we haven't covered, and briefly describe it to your fellow students, again with an interesting application if you can find one.

Categorical distribution

I found from Murphy that the categorical distribution can be used to represent a finite set of labels $y \in \{1, 2, 3, \dots, C\}$ and it generalises the Bernoulli distribution to $C > 2$ values. (2022, p.53).

Murphy gives the categorical distribution (the pmf) as follows:

$$\text{Cat}(y | \theta) = \prod_{c=1}^C \theta_c^{I(y=c)}$$

Where $p(y = c | \theta) = \theta_c$ meaning the probability of the outcome y is ends up being a specific class c , is the c -th entry of the vector θ . The indicator function acts like a switch that is 0 for all events except the one that occurred, so that the product picks out the specific θ_c .

This is very similar to Bernoulli but just with C factors instead of only two.

I found this video with an example around weather to be really helpful:
<https://www.youtube.com/watch?v=421uW9aZHio>

I read about classification distributions being very common across Machine Learning in Ruiz et al. (2018) with applications in classification, reinforcement learning, recommendation systems and neural attention models. They mention that the most common means to form a categorical is using softmax transformation. I have seen the softmax function come up quite a bit in my research and plan to research this further.

Beta distribution

I read about the Beta distribution in Murphy (2022, pp.63-64) where he states its probability density function (pdf) as follows:

$$\text{Beta}(x|a, b) = \frac{1}{B(a, b)} x^{a-1} (1 - x)^{b-1}$$

Where $B(a, b)$ is the beta function which is defined as:

$$B(a, b) \triangleq \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$$

Where $\Gamma(a)$ is the Gamma function which he states as:

$$\Gamma(a) \triangleq \int_0^{\infty} x^{a-1} e^{-x} dx$$

He also notes the a, b need to be greater than 0 for the density to be integrable.

He also gives the mean and variance as the following:

$$\text{mean} = \frac{a}{a+b}$$
$$\text{var} = \frac{ab}{(a+b)^2(a+b+1)}$$

Interestingly Murphy also notes that when $a = b = 1$ then it reduces to a uniform distribution.

I found this video called [The beta distribution: data science basics](https://www.youtube.com/watch?v=421uW9aZHio) really helpful to understand this distribution.

I did some research to see where beta distributions are applied in AI. I found that they can be directly applied to LLM benchmarking. Manousaridis et al. (2026) note that standard benchmarks usually use single point estimates that hide the variability in question difficulty and the non-deterministic nature of the model. To address this they introduce their hierarchical Bayesian Beta-Binomial framework. As part of this framework they aim to address three fundamental areas of LLM evaluation including the differences in question difficulty which is where the beta distribution is leveraged. They placed a Beta prior on the probability of each question being answered correctly. They tested 11 models using the MMLU question dataset and found that for some models rephrasing the questions had a significant impact on inter-question homogeneity (κ) i.e. making the question difficulty more homogeneous. This effect can be viewed in Figure 2a as an example for the Gemini Model 2.0 Flash.

References:

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